IoT-Based Industrial Telepresence Rover for Remote Inspection and Environmental Monitoring

Project Documentation

Application Domain: Paper and Pulp Industry / Pulp Mill Environment

A prototype-based embedded systems and IoT solution combining remote rover movement, live camera streaming, environmental sensing, obstacle warning, and remotely controlled water-pump actuation.

Primary controller	ESP32 DevKit V1 (38-pin)
Camera subsystem	Separate ESP32-CAM board
Connectivity	Wi-Fi, Blynk IoT dashboard, camera stream URL
Power architecture	2S Li-ion battery with regulated 6 V, 5 V and 3.3 V rails

Prepared as a working technical baseline. Confirm component labels, module variants, and current ratings before final wiring and testing.

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Important baseline: the two physical I²C LCDs and the mist maker have been removed. Sensor values are displayed in Blynk. The relay controls only the mini DC water pump. The ESP32-CAM runs independently from the main ESP32.

1. Project Overview

The IoT-Based Industrial Telepresence Rover is a small mobile prototype intended to demonstrate remote inspection and environmental monitoring concepts in a simulated paper-and-pulp mill environment. An ESP32 DevKit V1 handles movement commands, sensor acquisition, obstacle warning, buzzer operation and relay-based pump control. A separate ESP32-CAM provides live video over Wi-Fi. Blynk IoT is the remote dashboard for rover controls and sensor readings.

The system is teleoperated: the operator chooses the movement direction remotely. Obstacle sensing provides a near-field warning and can inhibit forward movement when the measured distance is at or below 15 cm. The MQ-135 is used for prototype-level sensor response/air-quality indication, not certified industrial gas measurement.

2. Problem Statement, Objectives and Scope

2.1 Problem statement

Inspection in industrial areas may expose personnel to heat, moisture, dust, restricted access, or other operating hazards. A compact telepresence rover can provide a remote visual viewpoint and basic environmental readings while demonstrating embedded control and IoT-based monitoring. This student prototype is a proof of concept, not a certified industrial safety device.

2.2 Objectives

• Provide remote forward, backward, left, right and stop control through Blynk.

• Stream live video using a separate ESP32-CAM subsystem.

• Measure temperature and relative humidity with a DHT22.

• Read MQ-135 sensor output as a raw ADC-based air-quality indicator.

• Measure nearby obstacle distance using an HC-SR04 ultrasonic sensor.

• Provide automatic obstacle warning and stop forward motion at a 15 cm threshold.

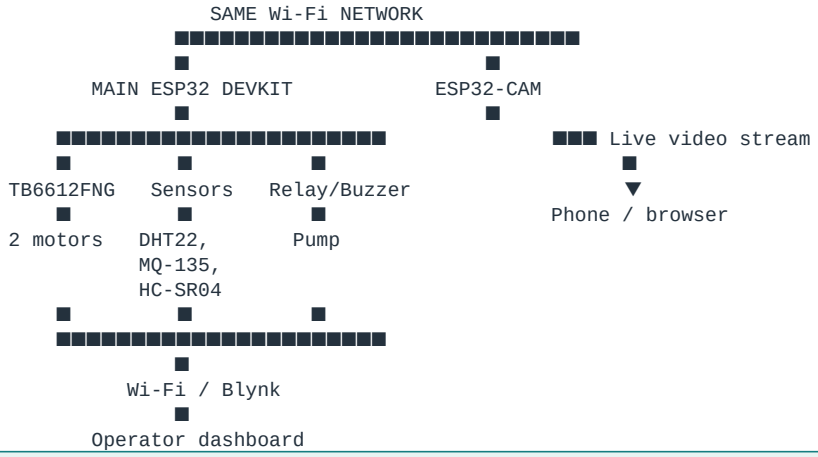
• Control an active buzzer and a mini DC water pump remotely.

• Use regulated power rails and verify electrical compatibility between sensors, driver and ESP32.

2.3 Scope and exclusions

Included: Wi-Fi teleoperation, Blynk display/control, sensor monitoring, obstacle warning, live camera streaming, relay-controlled pump actuation and manual prototype testing. Excluded: AI/ML, SLAM, LiDAR, ROS, computer vision, GPS navigation, autonomous path planning, robotic arm, camera pan servo, headlights, physical LCDs and mist maker.

3. System Architecture



The main ESP32 and ESP32-CAM are separate electronic subsystems with separate firmware. For the selected architecture, they do not require a direct GPIO link. They can be powered from the same battery-derived 5 V rail only if the supply has adequate current capacity and wiring; otherwise use a separate regulated 5 V branch. Their grounds should share a common reference when supplied from the same low-voltage system.

4. Hardware Components and Operating Voltages

Component	Nominal / operating supply	Connection / notes
2 × 18650 Li-ion cells, 2500 mAh each	3.6–3.7 V nominal per cell; 4.2 V full per cell	Series pack: 7.2–7.4 V nominal, 8.4 V fully charged; pack capacity remains 2500 mAh
2S BMS	For 2-series Li-ion cells	Match current rating and cell configuration; follow B-, BM/B1 and B+ markings
6 V buck converter	Input must cover 2S battery range; output 6.0 V	Drive rail; size for motor startup/stall current
LM2596 buck converter	Input from 2S pack; output set to 5.0 V	Electronics rail; verify thermal/current capability
ESP32 DevKit V1	5 V at 5V/VIN input; 3.3 V logic	3V3 pin is regulated output; don't apply 5 V to GPIO
ESP32-CAM	Typically 5 V at 5V pin	Separate camera subsystem; provide stable supply with transient-current margin
TB6612FNG	VCC logic: 2.7–5.5 V; VM motor supply: motor rail	VCC = 3.3 V; VM = regulated 6 V; common GND
2 × BO geared DC motors	6 V rated	Connect to TB6612FNG A01/A02 and B01/B02
DHT22	Approximately 3.3–6 V	Use 3.3 V; GPIO4 data; add pull-up if module lacks one
MQ-135 module	Common modules use 5 V heater/module supply	AOUT through divider/conditioner to GPIO34; raw ADC is not calibrated AQI
HC-SR04	Typically 5 V; commonly specified about 4.5–5.5 V	TRIG GPIO27; ECHO through divider/level shifter to GPIO35

Component	Nominal / operating supply	Connection / notes
1-channel relay module	Typically 5 V coil/module supply	IN = GPIO26 in this baseline; verify active level and 3.3 V input compatibility
Mini DC water pump	Must match pump label (not yet verified)	Switch with relay contacts; separate supply branch if needed
Active buzzer	Depends on actual module	GPIO13 through transistor/MOSFET driver if current exceeds safe GPIO drive
Physical 16×2 I²C LCDs	Not used	Removed; display sensor values in Blynk instead
Ultrasonic mist maker	Not used	Removed from final configuration

Voltage compatibility rule: ESP32 GPIOs are 3.3 V logic and are not 5 V tolerant. The HC-SR04 ECHO and MQ-135 AOUT must be checked and reduced to a safe ESP32 ADC/input voltage. The example 10 kΩ series/top resistor and 20 kΩ bottom resistor divider converts 5 V to about 3.33 V ideally; actual module output and resistor tolerances must be considered.

5. Power Distribution and Battery Design

5.1 Power rails

Rail	Source	Loads
Battery bus	2S pack through 2S BMS and master switch	Inputs to buck converters only; do not connect directly to 6 V motors or ESP32 5V pin
6.0 V drive rail	Dedicated 6 V buck converter	TB6612FNG VM only; driver outputs power the two 6 V motors
5.0 V electronics rail	LM2596 set to 5.0 V	ESP32 5V/VIN, MQ-135, HC-SR04, relay module, ESP32-CAM and pump only if pump is rated 5 V and rail capacity is adequate
3.3 V logic rail	ESP32 DevKit 3V3 output	TB6612FNG VCC and DHT22; only low-current peripherals within board regulator limits
Common ground	Ground reference shared between subsystems	Connect grounds; route motor/pump high-current returns away from sensitive signal return paths

5.2 Battery capacity and runtime

Two 2500 mAh cells in series form a 2S1P pack: approximately 7.4 V nominal and 2500 mAh capacity. The nominal energy is about 7.4 V × 2.5 Ah = 18.5 Wh. Series connection increases voltage, not amp-hour capacity. Runtime depends on actual average battery current, converter efficiency, motor load, pump current, camera consumption and battery condition.

```
Nominal pack energy ≈ Vnominal × Ah
                    ≈ 7.4 V × 2.5 Ah
                    ≈ 18.5 Wh

Estimated runtime (hours) ≈ usable battery energy (Wh) / average system power (W)
```

The BMS, cells, wiring, connectors and regulators must support peak motor startup/stall current plus the camera and actuator loads. Do not treat a cell's advertised capacity as proof that it can safely supply the required peak current. Use matched cells, a correctly rated 2S BMS and a suitable 8.4 V CC/CV charger.

6. Main ESP32 Circuit Connections

6.1 TB6612FNG and motors

TB6612FNG pin	Connect to
VM	Regulated 6.0 V drive rail
VCC	ESP32 3.3 V output

TB6612FNG pin	Connect to
GND	Common ground
PWMA	ESP32 GPIO25
AIN1 / AIN2	ESP32 GPIO16 / GPIO17
PWMB	ESP32 GPIO14
BIN1 / BIN2	ESP32 GPIO18 / GPIO19
STBY	ESP32 GPIO23; drive HIGH to enable
A01 / A02	Left motor terminals
B01 / B02	Right motor terminals

The 6 V rail is connected to VM only. VCC is the 3.3 V logic supply. Motor direction depends on physical motor orientation; if a wheel turns in the wrong direction, power down and reverse that motor's two output wires or adjust the direction logic.

6.2 DHT22

```
DHT22 VCC  -> ESP32 3V3
DHT22 GND  -> Common GND
DHT22 DATA -> ESP32 GPIO4
Pull-up    -> 4.7 kΩ to 10 kΩ from DATA to 3.3 V if not already present
```

6.3 MQ-135

```
MQ-135 VCC -> 5 V rail (common module type)
MQ-135 GND -> Common GND
MQ-135 AOUT -> voltage divider / signal conditioning -> ESP32 GPIO34
```

6.4 HC-SR04

```
HC-SR04 VCC  -> 5 V rail
HC-SR04 GND  -> Common GND
HC-SR04 TRIG -> ESP32 GPIO27
HC-SR04 ECHO -> voltage divider / level shifter -> ESP32 GPIO35
```

6.5 Relay and mini DC water pump

Control side: relay VCC → 5 V rail; relay GND → common GND; relay IN → ESP32 GPIO26. Verify that the specific relay module reliably accepts 3.3 V control logic. Many modules are active-LOW, but this must be tested rather than assumed.

```
Pump supply positive -> Relay COM
Relay NO             -> Pump positive
Pump negative        -> Pump supply negative / GND as appropriate

Use COM + NO so the pump is OFF when the relay is de-energized.
The pump supply voltage must match the pump label.
```

The relay contact switches pump current; the ESP32 GPIO does not power the pump. Verify contact current rating and pump startup/stall current. If the pump is an inductive DC load, use suitable suppression where appropriate and follow the relay module manufacturer's guidance.

6.6 Buzzer

```
ESP32 GPIO13 -> transistor/MOSFET driver input
Driver output -> active buzzer
Buzzer supply and GND -> match the buzzer rating
```

Use a transistor/MOSFET driver if the buzzer current is above a safe GPIO load. If the buzzer is a magnetic/inductive type, provide appropriate flyback protection according to its circuit.

7. Separate ESP32-CAM Streaming Circuit

```
Regulated 5 V supply -> ESP32-CAM 5V pin
Supply GND           -> ESP32-CAM GND
Camera               -> onboard camera connector
```

```
ESP32-CAM joins the same Wi-Fi network and runs its own CameraWebServer firmware.
The serial monitor reports the stream/web address after connection.
```

The ESP32-CAM is not programmed as part of the main ESP32 sketch. For a common AI-Thinker board, select the matching camera model in the Arduino CameraWebServer example. Confirm the actual board and camera module before selecting the model definition. Provide a stable 5 V supply with adequate peak-current capacity; camera and Wi-Fi activity can create current transients.

The main controller and camera communicate through the Wi-Fi network at the application level; no direct GPIO connection is required for basic live streaming. A stream URL can be opened separately or placed in a suitable dashboard link/widget if supported by the Blynk interface being used.

8. GPIO Allocation

ESP32 GPIO	Function	Direction / notes
GPIO4	DHT22 DATA	Digital input/output timing
GPIO13	Buzzer driver	Output
GPIO14	TB6612 PWMB	PWM output
GPIO16	TB6612 AIN1	Output
GPIO17	TB6612 AIN2	Output
GPIO18	TB6612 BIN1	Output
GPIO19	TB6612 BIN2	Output
GPIO21	Unused / available	No physical LCDs
GPIO22	Unused / available	No physical LCDs
GPIO23	TB6612 STBY	Output; HIGH enables driver
GPIO25	TB6612 PWMA	PWM output
GPIO26	1-channel relay / pump	Output
GPIO27	HC-SR04 TRIG	Output
GPIO34	MQ-135 AOUT	Input-only ADC; use voltage divider
GPIO35	HC-SR04 ECHO	Input-only; use voltage divider

9. Blynk IoT Datastream Configuration

Virtual pin	Name	Type	Range / meaning
V0	Front	Integer	0–1; 1 = command active
V1	Back	Integer	0–1; 1 = command active
V2	Left	Integer	0–1; 1 = command active
V3	Right	Integer	0–1; 1 = command active
V4	Stop	Integer	0–1; 1 = stop command
V5	Temperature	Double	0–80 °C display range
V6	Humidity	Integer	0–100 %
V7	Air Quality	Integer	0–4095 raw ESP32 ADC count; not calibrated AQI
V8	Obstacle Distance	Double	0–400 cm

Virtual pin	Name	Type	Range / meaning
V9	Obstacle	Integer	0 = clear; 1 = detected
V10	Buzzer	Integer	0 = OFF; 1 = ON
V11	Water Pump	Integer	0 = OFF; 1 = ON

The V-number is the datastream identifier used by the firmware. The internal datastream ID shown separately in the Blynk console is not used in Blynk.virtualWrite(Vx, ...) or BLYNK_WRITE(Vx). Configure V0–V4 as momentary/push movement buttons with ON=1 and OFF=0; this allows release to stop the rover when the firmware implements that behavior.

10. Software Logic and Operating Sequence

• Power the main ESP32 and ESP32-CAM from stable regulated supplies and join both boards to the same Wi-Fi network.

• The main ESP32 connects to Blynk and initializes the motor driver, DHT22, MQ-135, HC-SR04, buzzer and relay output.

• The operator presses Front, Back, Left, Right or Stop in the Blynk dashboard.

• The ESP32 sets TB6612FNG direction inputs and PWM outputs to drive the two motors.

• The DHT22 is sampled periodically and temperature/humidity values are sent to V5/V6.

• The MQ-135 analog output is sampled and the raw ADC value is sent to V7.

• The HC-SR04 distance is measured and sent to V8; V9 indicates clear/detected using the 15 cm threshold.

• If an obstacle is detected at or below 15 cm, the buzzer can sound and forward movement should be inhibited/stopped.

• V10 controls manual buzzer operation; the buzzer output is the logical OR of manual buzzer request and obstacle alert.

• V11 controls the relay, which switches the mini water pump using COM and NO contacts.

• The ESP32-CAM streams video independently; the operator opens its stream address while controlling the rover.

10.1 Simplified obstacle decision

```
Measure HC-SR04 distance
        |
        v
Is distance <= 15 cm?
    /             \
  YES              NO
   |                |
Obstacle=1       Obstacle=0
Buzzer alert     Clear status
Stop forward     Permit forward command
   |                |
  Send V9 status and V8 distance to Blynk
```

A timeout or invalid ultrasonic reading should not be interpreted as proof that the path is clear. For a robust implementation, define a sensor-fault state and stop forward motion if repeated readings are invalid. The example behavior in a starter sketch may require this refinement before unattended testing.

11. Safety, Limitations and Verification Notes

• Never connect the 7.4–8.4 V battery bus directly to the 6 V motors, TB6612FNG VM, or ESP32 5V/VIN pin.

• Connect the 6 V drive rail to TB6612FNG VM only; connect VCC to 3.3 V logic.

• All subsystems require a common ground reference, but high-current motor/pump returns should not pass through ESP32 signal-ground wiring.

• ESP32 GPIOs are not 5 V tolerant. Level-shift HC-SR04 ECHO and verify/divide MQ-135 AOUT before GPIO35/GPIO34.

• Confirm pump voltage, running current and startup/stall current from its label or measurement before finalizing the supply.

• Confirm relay input polarity and 3.3 V compatibility. Test pump OFF at boot and during Wi-Fi/Blynk disconnection.

• Use a 2S BMS with appropriate continuous and peak current ratings, matched cells, secure holders/welded connections and an 8.4 V charger.

• Set both converters with a multimeter before connecting electronics. Check regulator heating under load.

• Add suitable decoupling near the motor driver and power-hungry modules. Keep motor wiring short and separated from sensor signal wires.

• MQ-135 requires heater warm-up and calibration; report raw ADC response unless a defensible calibration method is implemented.

• The HC-SR04 has blind zones, beam-angle limitations and possible false readings on angled/soft surfaces. It is not a safety-rated industrial sensor.

• Blynk/Wi-Fi latency or disconnection can interrupt commands. A robust firmware version should stop motors and turn the pump off when the control connection is lost.

12. Testing Plan and Observation Tables

Record actual measured results; do not fill a project report with invented values. Test each subsystem separately before full integration.

Test	Procedure	Record
Power rails	Measure battery, 6 V and 5 V outputs unloaded and under load	Battery V, drive V, electronics V, regulator temperature
Motor direction	Test each wheel briefly with chassis raised	Forward/reverse correctness, noise, driver temperature
PWM response	Test several PWM duty cycles on a safe, unloaded chassis	Duty %, RPM, motor current
Obstacle threshold	Place a flat target at measured distances around 15 cm	Actual distance, reported distance, status, stop response
DHT22	Compare with a reference thermometer/hygrometer if available	Temperature and humidity readings
MQ-135	Allow recommended warm-up and record readings in stable conditions	Raw ADC count and test conditions
Blynk response	Toggle each control and observe device response	Command latency, missed commands, reconnect behavior
Pump/relay	Switch pump repeatedly with water path secured	Pump voltage/current, relay response, leakage, heat
Camera stream	Stream at intended resolution over Wi-Fi	Startup time, frame continuity, disconnections
Battery endurance	Run representative duty cycle and log pack voltage/time	Runtime, average current/power, low-voltage behavior

12.1 Observation table templates

PWM duty (%)	Left RPM	Right RPM	Battery current (A)	Notes
25				

PWM duty (%)	Left RPM	Right RPM	Battery current (A)	Notes
40				
55				
70				
85				

Target distance (cm)	Measured distance (cm)	V9 status	Buzzer	Forward stop?
10				
15				
20				
30				

Elapsed time (min)	Battery voltage (V)	System current (A)	Operating state
0			Startup
10			Driving / sensing
20			Driving / sensing
30			Driving / sensing

13. Conclusion

The proposed rover integrates embedded motor control, Wi-Fi teleoperation, environmental sensing, obstacle warning, relay-controlled water-pump actuation and independent live video streaming. The three-rail power architecture separates 6 V motor power from 5 V peripherals and 3.3 V logic. Blynk provides a single remote interface for movement, sensor values, buzzer and pump control, while the ESP32-CAM provides the visual channel. Before final demonstration, the team should verify all supply voltages, signal-level protection, pump/relay ratings, battery current capability, Wi-Fi fail-safe behavior and measured performance. Results and claims should be based on recorded experiments.

Appendix A — Pre-Power Checklist

- ■ Battery polarity, 2S BMS wiring and fuse/protection arrangement verified.
- ■ 6 V buck output checked with multimeter; connected only to TB6612FNG VM.
- ■ LM2596 output checked and set to 5.0 V before powering electronics.
- ■ ESP32 3V3 rail used only within the DevKit regulator's current capability.
- ■ Common ground established; motor and pump current returns routed appropriately.
- ■ MQ-135 AOUT and HC-SR04 ECHO cannot exceed safe ESP32 input voltage.
- ■ Relay default state is pump OFF; COM/NO wiring verified.
- ■ Pump voltage and current rating verified from the actual component.
- ■ Motor chassis lifted for initial direction/PWM testing.
- ■ ESP32-CAM supply and firmware tested independently.
- ■ Blynk datastreams V0–V11 match the firmware exactly.
- ■ Emergency power switch is accessible during testing.